MAT313: Differential Geometry for Curves and Surfaces

Mahmudul Hasan Turjoy

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To be written...

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1 Lecture 1

§1.1 Background discussion

A **parametrized differentiable curve** is a differentiable map

$$\alpha : I \to \mathbb{R}^3 \tag{1.1}$$

of an open interval $I = (a, b)$ of $\mathbb{R}$ into $\mathbb{R}^3$.

Example 1.1.1

The map

$$\alpha : \mathbb{R} \to \mathbb{R}^3 \quad \text{by} \quad \alpha(t) = (t^3, t^2), \quad \text{where} \quad t \in \mathbb{R} \tag{1.2}$$

is a parametrized differentiable curve. Here, $x = t^3$ and $y = t^2$. Eliminate t to obtain $y^3 = x^2$. Therefore, the curve has the following trace:

In this example, $\alpha'(t) = (3t^2, 2t)$. At $t = t_0 \neq 0$, the slope of the tangent line is $m = \frac{2t_0}{3t_0^2} = \frac{2}{3t_0}$. And the tangent line goes through $\alpha(t_0) = (t_0^3, t_0^2)$. Hence, the equation of the tangent at $t = t_0$ is

$$y - t_0^2 = \frac{2}{3t_0}\left(x - t_0^3\right). \tag{1.3}$$

Add graph

Definition 1.1.2. Let $\alpha : I \to \mathbb{R}^3$ be a parametrized differentiable curve. For each $t \in I$ where $\alpha'(t) \neq 0$, there is a well-defined straight line which goes through $\alpha(t)$ and contains the vector $\alpha'(t)$. This line is called the **tangent line to α at t**. Any point t where $\alpha'(t) = 0$ is a **singular point** of α.

In example Example 1.1.1, $t = 0$ is a singular point. We restrict our attention to curves without any singular points.

Definition 1.1.3 (Regular curve)**.** A parametrized differentiable curve $\alpha : I \to \mathbb{R}^3$ is said to be **regular** if $\alpha'(t) \neq 0$ for all $t \in I$.

Given $t \in I$, the arc length of a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ from the point t_0 is

$$s(t) = \int_{t_0}^{t} |\alpha'(u)| du, \tag{1.4}$$

where

$$|\alpha'(u)| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} \tag{1.5}$$

given that $\alpha(u) = (x(u), y(u), z(u))$, and so $\alpha'(u) = (x'(u), y'(u), z'(u))$.

The arc length $s(t)$ is a differentiable function of t. Indeed, $|\alpha'| = |\cdot| \circ \alpha'$ is continuous, for being the composition of two continuous functions. Therefore, by the First Fundamental Theorem of Calculus,

$$s(t) = \int_{t_0}^{t} |\alpha'(u)| du \tag{1.6}$$

$$\implies s'(t) = |\alpha'(t)|. \tag{1.7}$$

It may well be the case that the parameter t itself is the arch length s, i.e., $s = t$. Then $\frac{ds}{dt} = 1 = |\alpha'(t)|$. And so

$$s = \int_{t_0}^{t} |\alpha'(u)|du = \int_{t_0}^{t} 1du = t - t_0. \tag{1.8}$$

§1.2 The local theory of curves parametrized by arc length

Let $\alpha : I = (a, b) \to \mathbb{R}^3$ be a curve parametrized by arc length s. We know that the speed is $|\alpha'(s)| = 1$. Hence, the acceleration is normal[1] to the tangent vector at the point. This acceleration vector $\alpha''(s)$ pulls the object and deviates it from the tangent line. If it were zero, the object would have travelled along the tangent line. The more the acceleration, i.e., the more $|\alpha''(s)|$ is, the more the deviation of the trajectory of the object or curve from the tangent line at s.

Definition 1.2.1 (Curvature)**.** Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length $s \in I$. The number

$$\kappa(s) \coloneqq |\alpha''(s)| \tag{1.9}$$

is called the **curvature of** α at s.

Orientation a la do Carmo. The ambient space where the curves and surfaces are living in is $\mathbb{R}^3$. The curve $\alpha(s)$ living in $\mathbb{R}^3$ has a velocity vector

$$\alpha'(s) = \begin{bmatrix} x'(s) \\ y'(x) \\ z'(s) \end{bmatrix}. \tag{1.10}$$

To discuss the orientation of this curve, do Carmo reverses the direction of traversal as the parameter of the curve increase.

add picture

Recall the curvature $\kappa(s)$ of α at s, given by Equation 1.9. At points where $\kappa(s) \neq 0$, a unit vector $n(s)$ in the direction of $\alpha'(s)$ is well-defined by the equation

$$\alpha''(s) = \kappa(s) \cdot n(s). \tag{1.11}$$

From $\alpha'(s) \cdot \alpha'(s) = 1$, one obtains by differentiation $\alpha''(s) \cdot \alpha'(s) = 0$, and from Equation 1.11 one finds that $n(s)$ is normal to $\alpha'(s)$, i.e.,

$$n(s) \cdot \alpha'(s) = \frac{\alpha''(s)}{\kappa(s)} \alpha'(s) = 0. \tag{1.12}$$

$n(s)$ is called the **normal vector at** s.

The plane determined by the unit tangent vector $\alpha'(s)$ and the normal vector $n(s)$ is called the **osculating plane**.

Definition 1.2.2 (Singular points of order 0 and 1)**.** For the arc-parametrized curve $\alpha(s)$, s is a **singular point of order** 0 if $\alpha'(s) = 0$ and it is a **singular point of order** 1 if $\alpha''(s) = 0$ with $\alpha'(s) \neq 0$.

[1] $\alpha(s) \cdot \alpha(s) = 1 \implies 2\alpha''(s) \cdot \alpha'(s) = 0 \implies \alpha''(s) \perp \alpha'(s)$